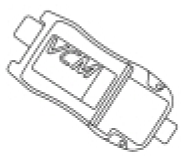
Instrument Cluster and Panel Illumination

Ford Ecat Electronic Parts Catalog

Special Tool(s)

 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent

Principles of Operation

Dimmable Backlighting

When the parking lamps are on, the Steering Column Control Module (SCCM) monitors the input from the instrument panel dimmer switch. The SCCM sends a message to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) relaying the instrument panel dimmer switch input. Based on the input from the instrument panel dimmer switch, the BCM sends voltage to the dimmable switches. The BCM sends a message to the HVAC module and to the ACM (except base AM/FM) on the MS-CAN and a message to the Trailer Brake Control (TBC) module and Instrument Panel Cluster (IPC) over the High Speed Controller Area Network (HS-CAN) to indicate the backlighting intensity level. The IPC sends a message to indicate the backlighting intensity level to the APIM , FCIM , and FCDIM over the Infotainment Controller Area Network (I-CAN).

If the receiving module receives invalid backlighting data from the BCM or from the IPC for 5 seconds or less, the receiving module defaults the backlighting to the last setting. If the receiving module does not receive the backlighting status message from the BCM or from the IPC or if the data received is deemed invalid for more than 5 seconds, the receiving module sets a missing message related DTC in continuous memory and defaults the backlighting to full nighttime intensity.

Non-Dimmable Backlighting

When the accessory delay relay is energized, switched voltage is supplied to the rear heated seat switches, window control and door lock control switches.

Field-Effect Transistor (FET) Protection

A Field-Effect Transistor (FET) is a type of transistor that, when used with module software, monitors and controls current flow on module outputs. The FET protection strategy prevents module damage in the event of excessive current flow.

The BCM utilizes an FET protective circuit strategy for many of its outputs (for example, a headlamp output circuit). Output loads (current level) are monitored for excessive current (typically short circuits) and are shut down (turns off the voltage or ground provided by the module) when a fault event is detected. A short circuit DTC is stored at the fault event and a cumulative counter is started.

When the demand for the output is no longer present, the module resets the FET protection to allow the circuit to function. The next time the driver requests a circuit to activate that has been shut down by a previous short (FET protection) and the circuit is still shorted, the FET protection shuts off the circuit again and the cumulative counter advances.

When the excessive circuit load occurs often enough, the module shuts down the output until a repair procedure is carried out. Each FET protected circuit has 3 predefined levels of short circuit tolerance based on the harmful effect of each circuit fault on the FET and the ability of the FET to withstand it. A module lifetime level of fault events is established based upon the durability of the FET . If the total tolerance level is determined to be 600 fault events, the 3 predefined levels would be 200, 400 and 600 fault events.

When each tolerance level is reached, the short circuit DTC that was stored on the first failure cannot be cleared by the clear the continuous DTCs command. The module does not allow this code to be cleared or the circuit restored to normal operation until a successful self-test proves the fault has been repaired. After the self-test has successfully completed (no on-demand DTCs present), DTC U1000:00 and the associated DTC (the DTC related to the shorted circuit) automatically clears and the circuit function returns.

When each level is reached, the DTC associated with the short circuit sets along with DTC U1000:00. These DTCs are cleared using the module on-demand self-test, then the Clear DTC operation on the scan tool (if the on-demand test shows the fault corrected). The module never resets the fault event counter to zero and continues to advance the fault event counter as short circuit fault events occur.

If the number of short circuit fault events reach the third level, then DTCs U1000:00 and U3000:49 set along with the associated short circuit DTC. DTC U3000:49 cannot be cleared and the module must be replaced after the repair.

The BCM FET protected output circuit for the instrument cluster and panel illumination system is the dimmable switch output circuit.

Inspection and Verification

1. Verify the customer concern.

2. Visually inspect for obvious signs of electrical damage.

Visual Inspection Chart

Electrical
• Body Control Module (BCM) fuse(s): ▪ 12 (15A) ▪ 32 (15A) • Wiring, terminals or connectors • Illuminated components • BCM

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

4. **NOTE:** *Make sure to use the latest scan tool release software*.

   If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

5. **NOTE:** *The Vehicle Communication Module (VCM) prove-out confirms power and ground from the DLC are provided to the VCM .*

   If the scan tool does not communicate with the VCM :
   - check the VCM connection to the vehicle.
   - check the scan tool connection to the VCM .
   - refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:
   - verify the ignition key is in the ON position.
   - verify the scan tool operation with a known good vehicle.
   - refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.

7. Carry out the network test.
   - If the scan tool responds with no communication with one or more modules, refer to Section 418-00.
   - If the network test passes, retrieve and record the continuous memory DTCs.

8. Clear the continuous DTCs and carry out the self-test diagnostics for the BCM .

9. If the DTCs retrieved are related to the concern, go to DTC Charts. For all other DTCs, refer to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.
   If no DTCs related to the concern are retrieved, GO to Symptom Chart.

DTC Charts

Steering Column Control Module (SCCM) DTC Chart

DTC	Description	Action
B1124:09	Lamp Fade Control: Component Failure	GO to Pinpoint Test A.
All other DTCs	—	REFER to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.

Body Control Module (BCM) DTC Chart

DTC	Description	Action
B1315:11	Backlighting (Non Reflective Controls) Illumination Output: Circuit Short To Ground	GO to Pinpoint Test B.
B1315:15	Backlighting (Non Reflective Controls) Illumination Output: Circuit Short To Battery or Open	If the backlighting components are inoperative, GO to Pinpoint Test B. If the backlighting components are always on, GO to Pinpoint Test C.
U1000:00	Solid State Driver Protection Active - Driver Disabled: No Sub Type Information	The module has temporarily disabled an output because an excessive current draw exists (such as a short to ground). The BCM cannot enable the output until the cause of the short is corrected. ADDRESS all other DTCs first. After the cause of the concern is corrected, CLEAR the DTCs. REPEAT the self-test.

DTC	Description	Action
U3000:49	Control Module: Internal Electrical Failure	The module has permanently disabled an output because an excessive current draw fault (such as a short to ground) has exceeded the limits that the BCM can withstand. CORRECT the cause of the excessive current draw before installing a new BCM . ADDRESS all other DTCs first. After the cause of the concern is corrected, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
All other DTCs	—	REFER to the Diagnostic Trouble Code (DTC) Chart in Section 419-10.

FCIM DTC Chart

DTC	Description	Action
B113C:11	Hazard Switch Illumination: Circuit Short To Ground	CLEAR the DTC. WAIT at least 10 seconds. REPEAT the self-test. If DTC B113C:11 is retrieved again, INSTALL a new FCIM . Refer to the appropriate section in Group 415 for the procedure. TEST the system for normal operation.
All other DTCs	—	Refer to the appropriate section in Group 415 for the procedure.

Symptom Chart

Symptom Chart

Condition	Possible Causes	Action
• All dimmable illumination does not dim	• Wiring, terminals or connectors • Instrument panel dimmer switch • Steering Column Control Module (SCCM)	• GO to Pinpoint Test A.
• All dimmable non-networked illumination is inoperative	• Fuse • Wiring, terminals or connectors • Body Control Module (BCM)	• GO to Pinpoint Test B.
• All dimmable non-networked illumination is always on	• Wiring, terminals or connectors • BCM	• GO to Pinpoint Test C.
• The steering wheel switch(es) illumination is	• Wiring, terminals or connectors	• GO to Pinpoint Test D.

inoperative	• Steering wheel switch(es) • Steering wheel harness • Clockspring	
• One or more illumination source is inoperative	• Wiring, terminals or connectors • Door lock control switch • Illuminated switch	• GO to Pinpoint Test E.
• The window control switch illumination is inoperative	• Window control switch	• CHECK the operation of the power windows. • If the power windows operate correctly, INSTALL a new window control switch. REFER to Section 501-11. TEST the system for normal operation. • If the power windows do not operate correctly, REFER to Section 501-11.
• All network controlled illumination does not dim	• Controller Area Network (CAN) concern • BCM	• CARRY OUT the network test. • If the scan tool responds with no communication to any module on the CAN , REFER to Section 418-00. • If the scan tool communicates with the modules on the CAN , INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
• The Instrument Panel Cluster (IPC) illumination is inoperative	• High Speed Controller Area Network (HS-CAN) • IPC	• CARRY OUT the network test. • If the scan tool responds with no communication with the IPC , REFER to Section 418-00. • If the scan tool communicates with the IPC , INSTALL a new IPC . REFER to Section 413-01. TEST the system for normal operation.
• The HVAC module illumination is inoperative (without 4.2- inch [107mm] or 8- inch [203mm] display)	• Medium Speed Controller Area Network (MS-CAN) • HVAC module	• CARRY OUT the network test. • If the scan tool responds with no communication with the HVAC module, REFER to Section 418-00.

		• If the scan tool communicates with the HVAC module, INSTALL a new HVAC module. REFER to Section 412-01. TEST the system for normal operation.
• The Audio Front Control Module (ACM) illumination is inoperative (base AM/FM radio)	• Wiring, terminals or connectors • ACM	• GO to Pinpoint Test E.
• The Front Controls Interface Module (FCIM) illumination is inoperative/does not dim	• Medium Speed Controller Area Network (MS-CAN) (without 4.2- inch [107mm] display or 8- inch [203mm] touch screen) • Infotainment Controller Area Network (I-CAN) (with 4.2- inch [107mm] display or 8- inch [203mm] touch screen) • FCIM	• CARRY OUT the network test. • If the scan tool responds with no communication with the FCIM , REFER to Section 418-00. • If the scan tool communicates with the FCIM , INSTALL a new FCIM . Refer to the appropriate section in Group 415 for the procedure.
• The Front Display Interface Module (FDIM) illumination is inoperative/does not dim (non-touchscreen)	• MS-CAN • FDIM	• CARRY OUT the network test. • If the scan tool responds with no communication with the FDIM , REFER to Section 418-00. • If the scan tool communicates with the FDIM , INSTALL a new FDIM . REFER to Section 415-00B. TEST the system for normal operation.
• The FDIM illumination is inoperative/does not dim (with 8-inch [203mm] touchscreen)	• APIM • FDIM	• VERIFY that the audio system can be operated from the FDIM . • If the audio system operates correctly from the FDIM inputs, INSTALL a new APIM. Refer to the appropriate Removal and Installation procedure in Section 415-00. If the APIM replacement does not resolve the condition, replace the FDIM . REFER to Section 415-00D or Section 415-00E. TEST the system for normal operation. • If the audio system does not operate correctly from the FDIM inputs, REFER to Section 415-00D or Section 415-00E.

• The FCDIM illumination is inoperative/does not dim (with 4.2- inch [107mm] display)	• Infotainment Controller Area Network (I-CAN) • FCDIM	• CARRY OUT the network test. • If the scan tool communicates with the FCDIM , INSTALL a new FCDIM . REFFER to: Section 415-00C. TEST the system for normal operation. • If the scan tool does not communicate with the FCDIM , REFER to Section 418-00.
• The hazard switch/PAD indicator illumination is inoperative/does not dim (with 4.2- inch [107mm] and 8- inch [203mm] display only)	• FCIM	• INSTALL a new FCIM . Refer to the appropriate section in Group 415 for the procedure. TEST the system for normal operation.
• The hazard switch/PAD indicator/traction control switch illumination is inoperative/does not dim (without 4.2- inch [107mm] or 8- inch [203mm] display)	• Wiring, terminals or connectors • Illuminated switch	• GO to Pinpoint Test E.
• Stability/traction control switch illumination is inoperative/does not dim (with 4.2- inch [107mm] or 8- inch [203mm] display)	• Wiring, terminals or connectors • Illuminated switch	• GO to Pinpoint Test E.
• The Trailer Brake Control (TBC) module illumination is inoperative	• HS-CAN • TBC module	• CARRY OUT the network test. • If the scan tool responds with no communication with the TBC module, REFER to Section 418-00. • If the scan tool communicates with the TBC module, INSTALL a new TBC module, REFER to Section 206-10. TEST the system for normal operation.

Pinpoint Tests

Pinpoint Test A: All Dimmable Illumination Does Not Dim

Refer to Wiring Diagrams Cell 71 , Cluster and Panel Illumination for schematic and connector information.

Normal Operation

When the parking lamps are on, the dimmable backlighting intensity can be changed by pressing the instrument panel dimmer switch to the first detent up or first detent down. The instrument panel dimmer switch is a momentary contact switch. When the instrument panel dimmer switch is placed in the first detent up or down, the corresponding input signal is routed back to the SCCM to an internal ground indicating a request to turn the dimmable lighting up or down. The SCCM retains the last brightness setting between uses.

Based on the input received from the instrument panel dimmer switch, the SCCM sends a message to the Body Control Module (BCM) over the High Speed Controller Area Network (HS-CAN) . The BCM sends a network message to the backlit modules indicating the backlighting intensity level and a voltage to the hardwired dimmable illumination sources.

When the instrument panel dimmer switch is pressed to the second detent, the dome lamp request input circuit is routed to ground in addition to the instrument panel dimmer switch input circuit. The SCCM needs both an instrument panel dimmer switch up/down input circuit and the dome lamp request input circuit to turn the courtesy lamps on or off from the instrument panel dimmer switch.

If the SCCM detects a concern with the instrument panel dimmer switch up/down input circuit or the dome lamp request input circuit, the backlighting defaults to full nighttime brightness.

- DTC B1124:09 (Lamp Fade Control: Component Failure) — a continuous and on-demand DTC that sets when the SCCM detects a short to ground from the instrument panel dimmer switch up/down input or dome lamp request input circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Instrument panel dimmer switch
- SCCM

PINPOINT TEST A : ALL DIMMABLE ILLUMINATION DOES NOT DIM

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

A1 CHECK THE RECORDED RESULTS FROM THE SCCM ON-DEMAND SELF-TEST

- Ignition OFF.
- Review the recorded results from the SCCM on-demand self-test.

Is DTC B1124:09 present?

Yes	GO to A2.
No	GO to A4.

A2 CHECK THE INSTRUMENT PANEL DIMMER SWITCH

- Disconnect: Instrument Panel Dimmer Switch C2298.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: SCCM Self-Test .
- Clear the DTCs and repeat the SCCM self-test.

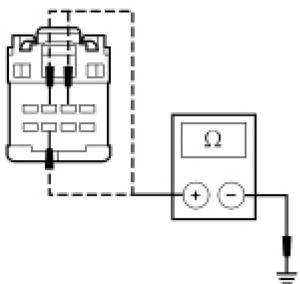
Is DTC B1124:09 still present?

Yes	GO to A3.
No	INSTALL a new instrument panel dimmer switch. REFER to Instrument Panel Dimmer Switch in this section. TEST the system for normal operation.

A3 CHECK THE INSTRUMENT PANEL DIMMER SWITCH INPUT CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: SCCM C2414A .
- Measure the resistance between the instrument panel dimmer switch, harness side and ground as follows:

Connector-Pin	Circuit
C2298 Pin 2	VLN36 (GY/OG)
C2298 Pin 3	CLN55 (YE/VT)
C2298 Pin 7	CLN56 (BN/VT)

- 

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Are the resistances greater than 10,000 ohms?

Yes	GO to A9.
No	REPAIR the circuit in question for a short to ground. CLEAR the DTCs. REPEAT the self-test.

A4 CHECK THE INSTRUMENT PANEL DIMMER SWITCH

- Disconnect: Instrument Panel Dimmer Switch C2298.
- Carry out the instrument panel dimmer switch component test. Refer to Wiring Diagrams Cell 149 for component testing.

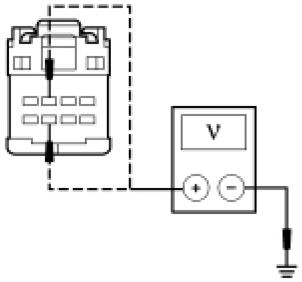
Is the instrument panel dimmer switch OK?

Yes	GO to A5.
No	INSTALL a new instrument panel dimmer switch. REFER to Instrument Panel Dimmer Switch in this section. TEST the system for normal operation.

A5 CHECK THE INSTRUMENT PANEL DIMMER SWITCH UP AND DOWN INPUT CIRCUITS FOR VOLTAGE

- Ignition ON.

- Measure the voltage between the instrument panel dimmer switch C2298 Pin 7, circuit CLN56 (BN/VT), harness side and ground; and between the instrument panel dimmer switch C2298 Pin 3, circuit CLN55 (YE/VT), harness side and ground.

- 
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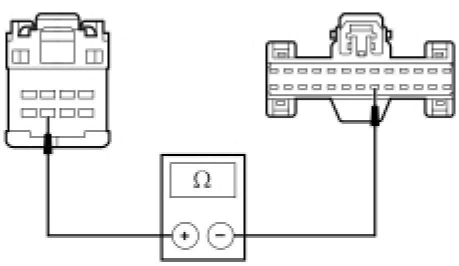
Are the voltages greater than 5 volts?

Yes	GO to A7.
No	GO to A6.

A6 CHECK THE INSTRUMENT PANEL DIMMER SWITCH UP AND DOWN INPUT CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: SCCM C2414A .
- Measure the resistance between the instrument panel dimmer switch C2298 Pin 7, circuit CLN56 (BN/VT), harness side and the SCCM C2414A Pin 19, circuit CLN56 (BN/VT), harness side.

- 
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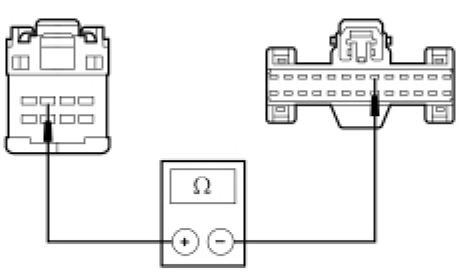
- Measure the resistance between the instrument panel dimmer switch C2298 Pin 3, circuit CLN55 (YE/VT), harness side and the SCCM C2414A Pin 6, circuit CLN55 (YE/VT), harness side.

- 
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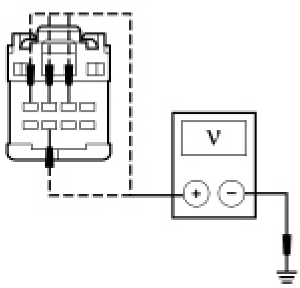
Are the resistances less than 5 ohms?

Yes	GO to A9.
No	REPAIR the circuit in question for an open. TEST the system for normal operation.

A7 CHECK THE INSTRUMENT PANEL DIMMER SWITCH INPUT AND RETURN CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: SCCM C2414A .
- Ignition ON.
- Measure the voltage between the instrument panel dimmer switch, harness side and ground as follows:

Connector-Pin	Circuit
C2298 Pin 2	VLN36 (GY/OG)
C2298 Pin 3	CLN55 (YE/VT)
C2298 Pin 4	RLN29 (WH/OG)
C2298 Pin 7	CLN56 (BN/VT)

- 

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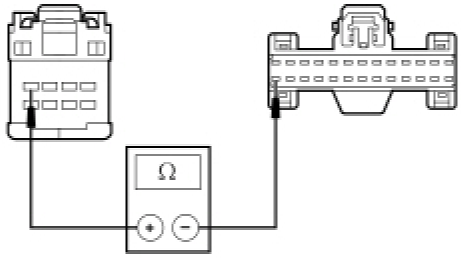
Is any voltage present?

Yes	REPAIR the circuit in question for a short to voltage. TEST the system for normal operation.
No	GO to A8.

A8 CHECK THE INSTRUMENT PANEL DIMMER SWITCH RETURN CIRCUIT FOR AN OPEN

- Ignition OFF.
- Measure the resistance between the instrument panel dimmer switch C2298 Pin 4, circuit RLN29 (WH/OG), harness side and the SCCM C2414A Pin 26, circuit RLN29 (WH/OG), harness side.

- 

Is the resistance less than 5 ohms?

Yes	GO to A9.
No	REPAIR the circuit for an open. TEST the system for normal operation.

A9 CHECK FOR CORRECT SCCM OPERATION

- Connect all disconnected connectors.
- Disconnect all the SCCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the SCCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new SCCM . REFER to Section 211-05. TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. RUN the on-demand self-test (required to clear certain DTCs). CORRECT any unresolved DTCs. CLEAR all DTCs. TEST the system for normal operation.

Pinpoint Test B: All Dimmable Non-Networked Illumination Is Inoperative

Refer to Wiring Diagrams Cell 71 , Cluster and Panel Illumination for schematic and connector information.

Normal Operation

When the parking lamps are on, the Body Control Module (BCM) provides voltage to the hardwired dimmable illumination sources based on the Steering Column Control Module (SCCM) illumination messages sent to the BCM over the network.

DTC Description	Fault Trigger Conditions
• B1315:11 — Backlighting (Non Reflective Controls) Illumination Output: Circuit Short To Ground	A continuous and on-demand DTC that sets when the BCM detects a short to ground from the backlighting output circuit. This DTC also sets when fuse 12 (15A) has failed or is removed.
• B1315:15 — Backlighting (Non Reflective Controls) Illumination Output: Circuit Short To Battery or Open	An on-demand DTC that sets when the BCM detects an open from the backlighting output circuit.
• U1000:00 — Solid State Driver Protection Active - Driver Disabled: No Sub Type Information	A continuous DTC that sets when the BCM has temporarily shut down the output driver. The module has temporarily disabled the backlighting output because an excessive current draw exists (such as a short to ground). The BCM cannot enable the backlighting output until the cause of the short is corrected, the DTCs have been cleared and a successful self-test is run.
• U3000:49 — Control Module: Internal Electrical Failure	A continuous DTC that sets when the BCM has permanently shut down the output driver. The module has permanently disabled the backlighting output because an excessive current draw fault (such as a short to ground) has exceeded the limits that the BCM can withstand. Correct the cause of the excessive current draw before installing a new BCM .

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- BCM

PINPOINT TEST B : ALL DIMMABLE NON-NETWORKED ILLUMINATION IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

B1 CHECK FOR RECORDED BCM DTCS FROM THE ON-DEMAND SELF-TEST

- Use the recorded results from the BCM self-test.

Is DTC B1315:11 present?

Yes	VERIFY the BCM fuse 12 (15A) is OK. If OK, GO to B2. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. After the repair: If no DTCs are present, TEST the system for normal operation. If DTC U1000:00 is present, CLEAR the DTCs **and** REPEAT the self-test **(required to enable the backlighting output driver if DTC U1000:00 is present).** TEST the system for normal operation. If DTC U3000:49 is present, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	GO to B2.

B2 BYPASS THE BCM

- Ignition OFF.
- Disconnect: BCM C2280B .

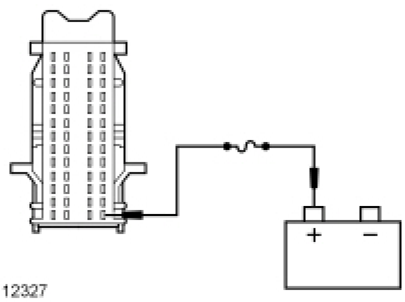
- **NOTE:** *If the jumper wire fails, refer to the Wiring Diagrams manual to identify the possible causes of the circuit short. After the repair:*

  If no DTCs are present, test the system for normal operation.

  If DTC U1000:00 is present, clear the DTCs ***and*** *repeat the self-test* ***(required to enable the courtesy lamp output driver if DTC U1000:00 is present).*** *Test the system for normal operation.*

  If DTC U3000:49 is present, INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . Test the system for normal operation.

  Connect a fused jumper wire between the BCM C2280B Pin 40, circuit VLN04 (VT/GY) and battery positive.

- 

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Do the backlighting components illuminate?

Yes	REMOVE the jumper wire. GO to B3.
No	REMOVE the jumper wire. REPAIR circuit VLN04 (VT/GY) for an open. TEST the system for normal operation.

B3 CHECK FOR CORRECT BCM OPERATION

- Disconnect all the BCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins
- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test C: All Dimmable Non-Networked Illumination Is Always On

Refer to Wiring Diagrams Cell 71 , Cluster and Panel Illumination for schematic and connector information.

Normal Operation

When the parking lamps are on, the Body Control Module (BCM) provides voltage to the hardwired dimmable illumination sources based on the Steering Column Control Module (SCCM) illumination messages sent to the BCM over the network.

- B1315:15 (Backlighting (Non Reflective Controls) Illumination Output: Circuit Short To Battery or Open) — an on-demand DTC that sets when the BCM detects a short to voltage from the backlighting output circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- BCM

PINPOINT TEST C : ALL DIMMABLE NON-NETWORKED ILLUMINATION IS ALWAYS ON

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

C1 CHECK THE EXTERIOR LIGHTING

- Place the headlamp switch in the OFF position.
- Observe the parking lights.

Are the parking lights illuminated?

Yes	REFER to Section 417-01.
No	GO to C2.

C2 CHECK THE BCM BACKLIGHTING OUTPUT CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: BCM C2280B .
- Ignition ON.

Do the dimmable backlighted components continue to illuminate?

Yes	REPAIR circuit VLN04 (VT/GY) for a short to voltage. TEST the system for normal operation.
No	GO to C3.

C3 CHECK FOR CORRECT BCM OPERATION

- Ignition OFF.
- Disconnect all the BCM connectors.
- Check for:
  - corrosion
  - damaged pins
  - pushed-out pins

- Connect all the BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector.

Pinpoint Test D: The Steering Wheel Switch(es) Illumination Is Inoperative

Refer to Wiring Diagrams Cell 71 , Cluster and Panel Illumination for schematic and connector information.

Normal Operation

When the parking lamps are on, the Body Control Module (BCM) sends voltage to the clockspring. The voltage passes through the clockspring to the steering wheel switches.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Steering wheel switch(es)
- Steering wheel harness
- Clockspring

PINPOINT TEST D : THE STEERING WHEEL SWITCH(ES) ILLUMINATION IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

D1 CHECK THE OTHER DIMMABLE LIGHTING SWITCHES

- Ignition ON.
- Place the headlamp switch in the PARKING LAMPS ON position.
- Press and hold the instrument panel dimmer switch up to the full illumination position and observe other dimmable switches.

Are all the dimmable steering wheel switches inoperative?

Yes	GO to Pinpoint Test A.
No	If all the steering wheel switches illumination is inoperative, GO to D2. If only one steering wheel switch illumination is inoperative, GO to D5.

D2 CHECK THE HORN OPERATION

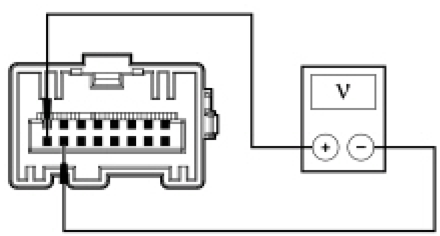
- Press the driver air bag module against the steering wheel.

Does the horn sound?

Yes	GO to D3.
No	REFER to Section 413-06 to diagnose the horn is inoperative.

D3 CHECK FOR VOLTAGE TO THE STEERING WHEEL SWITCHES

- Place the headlamp switch in the OFF position.
- Remove the driver air bag module. Refer to Section 501-20B.
- Disconnect: Clockspring C218B.
- Ignition ON.
- Place the headlamp switch in the PARKING LAMPS ON position.
- Press and hold the instrument panel dimmer switch up to the full illumination position.
- Measure the voltage between the clockspring C218B Pin 9, component side and the clockspring C218B Pin 10, component side.



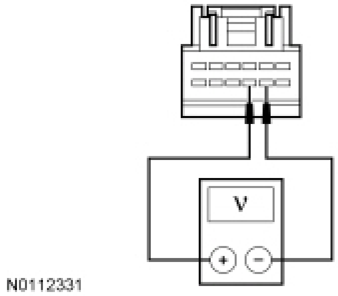
- N0116238

Is the voltage greater than 10 volts?

Yes	INSTALL a new steering wheel. REFER to Section 211-04. TEST the system for normal operation.
No	GO to D4.

D4 CHECK FOR VOLTAGE TO THE CLOCKSPRING

- Disconnect: Clockspring C218A.
- Measure the voltage between the clockspring C218A Pin 9, circuit VLN04 (VT/GY), harness side and the clockspring C218A Pin 8, circuit GD133 (BK), harness side.

- 

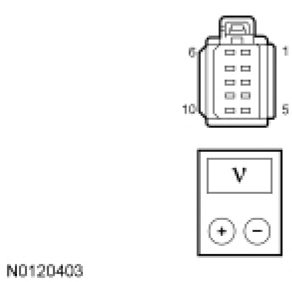
Is the voltage greater than 10 volts?

Yes	INSTALL a new clockspring. REFER to Section 501-20B. TEST the system for normal operation.
No	REPAIR circuit VLN04 (VT/GY) for an open. TEST the system for normal operation.

D5 CHECK THE SUSPECT STEERING WHEEL SWITCH FOR VOLTAGE

- Disconnect: Suspect Steering Wheel Switch .
- Press and hold the instrument panel dimmer switch up to the full illumination position.
- Measure the voltage between the inoperative illumination source, harness side as follows:

Component	Connector-Pin	Connector-Pin
Steering wheel switch — LH	C2998 Pin 2	C2998 Pin 4
Steering wheel switch — RH	C2999 Pin 1	C2999 Pin 3

- 

Is the voltage greater than 10 volts?

Yes	INSTALL a new component for the one in question. TEST the system for normal operation.
No	INSTALL a new steering wheel. REFER to Section 211-04. TEST the system for normal operation.

Pinpoint Test E: One Or More Illumination Source Is Inoperative

Refer to Wiring Diagrams Cell 71 , Cluster and Panel Illumination for schematic and connector information.

Normal Operation

The Body Control Module (BCM) sends voltage to the dimmable components.

For the base Audio Front Control Module (ACM) (AM/FM only), the BCM supplies voltage to the ACM illumination based on the Steering Column Control Module (SCCM) illumination messages sent to the BCM over the network.

When the accessory delay relay energizes, voltage is provided to the non-dimmable switches.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- ACM
- Door lock control switch
- Illuminated switch

PINPOINT TEST E : ONE OR MORE ILLUMINATION SOURCE IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

E1 CHECK THE VOLTAGE TO THE SINGLE ILLUMINATION

NOTE: *If the instrument panel dimmer switch illumination is the inoperative source, increase the backlighting brightness to the highest setting before unplugging the instrument panel dimmer switch connector.*

- Ignition OFF.
- Disconnect: Inoperative Illumination Source .
- Ignition ON.
- **NOTE:** *If the headlamp switch illumination is the inoperative source, unplugging the headlamp switch connector causes the BCM to turn the exterior lights on by default, so there is no need to turn the parking lamps on.*

  Place the headlamp switch to the PARKING LAMPS ON position.
- Increase the backlighting brightness to the highest setting.
- Measure the voltage between the inoperative illumination source, harness side and ground as follows:

Component	Connector-Pin	Circuit
ACM (base AM/FM radio only)	C290A Pin 3	VLN04 (VT/GY)
Floor shifter	C3245 Pin 4	VLN04 (VT/GY)
Hazard switch/PAD indicator/traction control switch (without 4.2- inch [107mm] or 8- inch [203mm] display)	C2355 Pin 6	VLN04 (VT/GY)
Stability/traction control switch (with 4.2- inch [107mm] or 8- inch [203mm] display)	C2114 Pin 5	VLN04 (VT/GY)
Headlamp switch	C205 Pin 1	VLN04 (VT/GY)

Component	Connector-Pin	Circuit
Hill descent control switch	C3394 Pin 3	VLN04 (VT/GY)
Instrument panel dimmer switch	C2298 Pin 1	VLN04 (VT/GY)
Mode select switch	C284 Pin 4	VLN04 (VT/GY)
Off-road mode switch (Raptor only)	C3393 Pin 3	VLN04 (VT/GY)
Overhead console (roof opening panel/power sliding window)	C912 Pin 2	VLN04 (VT/GY)
Upfitter switch	C3389 Pin 2	VLN04 (VT/GY)
LH rear heated seat switch	C731 Pin 2	CBP32 (GN/VT)
RH rear heated seat switch	C831 Pin 2	CBP32 (GN/VT)
LH door lock control switch	C505 Pin 1	CBP32 (GN/VT)
RH door lock control switch	C605 Pin 1	CBP32 (GN/VT)

Is the voltage greater than 10 volts?

Yes	For a door lock control switch, INSTALL a new door lock control switch. REFER to Section 501-14. TEST the system for normal operation. For rear heated seat switch, INSTALL a new rear heated seat switch. REFER to Section 501-10. For an instrument panel dimmer switch, INSTALL a new instrument panel dimmer switch. REFER to Instrument Panel Dimmer Switch in this section. TEST the system for normal operation. For all other components, GO to E2.
No	REPAIR circuit VLN04 (VT/GY) or circuit CBP32 (GN/VT) for an open. TEST the system for normal operation.

E2 CHECK THE GROUND CIRCUIT TO THE ILLUMINATION SOURCE FOR CONTINUITY

- Measure the voltage between the inoperative illumination source pins, harness side as follows:

Component	Connector-Pin/ Circuit	Connector-Pin/ Circuit
ACM (base radio only)	C290A Pin 3 VLN04 (VT/GY)	C290A Pin 4 GD114 (BK/BU)
Floor shifter	C3245 Pin 4 VLN04 (VT/GY)	C3245 Pin 5 GD138 (BK/WH)

Component	Connector-Pin/ Circuit	Connector-Pin/ Circuit
Hazard switch/PAD indicator/traction control switch (without 4.2- inch [107mm] or 8- inch [203mm] display)	C2355 Pin 6 VLN04 (VT/GY)	C2355 Pin 1 GD133 (BK)
Stability/traction control switch (with 4.2- inch [107mm] or 8- inch [203mm] display)	C2114 Pin 5 VLN04 (VT/GY)	C2114 Pin 3 GD133 (BK)
Headlamp switch	C205 Pin 1 VLN04 (VT/GY)	C205 Pin 8 GD133 (BK)
Hill descent control switch	C3394 Pin 3 VLN04 (VT/GY)	C3394 Pin 2 GD138 (BK/WH)
Mode select switch	C284 Pin 4 VLN04 (VT/GY)	C284 Pin 1 GD133 (BK)
Off-road mode switch (Raptor only)	C3393 Pin 3 VLN04 (VT/GY)	C3393 Pin 2 GD138 (BK/WH)
Overhead console (roof opening panel/power sliding window)	C912 Pin 2 VLN04 (VT/GY)	C912 Pin 12 CPR39 (VT/WH)
Upfitter switch	C3389 Pin 2 VLN04 (VT/GY)	C3389 Pin 3 GD138 (BK/WH)

Is the voltage greater than 10 volts?

Yes	INSTALL a new component for the one in question. TEST the system for normal operation.
No	REPAIR the ground circuit in question for an open. TEST the system for normal operation.